

Credit Card Approval Prediction

Phase 2: Requirement Analysis

SmartBridge / SkillWallet Virtual Internship Program

Team Member	Role
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1. Functional Requirements

ID	Requirement
FR-1	The system shall accept applicant demographic details (gender, family status, education, housing type).
FR-2	The system shall accept applicant financial details (annual income, income type, employment duration).
FR-3	The system shall accept applicant credit history summary (EMI paid off, EMI pastdues, number of loans).
FR-4	The system shall validate all required fields before submitting for prediction.
FR-5	The system shall preprocess submitted input identically to the training pipeline (encoding, scaling).
FR-6	The system shall return a binary prediction (Approved / Rejected) with a confidence percentage.
FR-7	The system shall support training and comparing multiple ML algorithms (Logistic Regression, Decision Tree, Random Forest).
FR-8	The system shall automatically select and persist the best-performing model for deployment.
FR-9	The system shall expose the trained model through a Flask web application.

2. Non-Functional Requirements

ID	Requirement
NFR-1	Performance: predictions shall return within 2 seconds under normal load.

NFR-2	Usability: the web form shall be simple enough for a non-technical analyst to use without training.
NFR-3	Reliability: the model shall be evaluated on a held-out test set with F1-score ≥ 0.85 before deployment.
NFR-4	Maintainability: preprocessing, training, and serving code shall be modular and independently testable.
NFR-5	Portability: the application shall run locally and be deployable to IBM Cloud without code changes.
NFR-6	Security: no personally identifiable applicant data shall be logged or persisted beyond the prediction.

3. Data Requirements

- **application_record.csv** — one row per applicant: gender, car/realty ownership, children count, income, income type, education, family status, housing type, age (DAYS_BIRTH), employment length (DAYS_EMPLOYED), occupation, family size.
- **credit_record.csv** — one row per applicant per month: payment status code (on-time, overdue by various amounts, paid off, or no active loan).
- Both files must share a common applicant ID column to allow merging.

4. User Stories

As a...	I want to...	So that...
Credit Analyst	enter an applicant's details and get an instant prediction	Instantly identify applications needing manual review.
Compliance Officer	batch-screen applicants with past-due records	Identify high-risk applicants for further review.
Prospective Customer	check my own eligibility before applying	Can avoid an unnecessary rejection on my credit file.
Data Scientist	compare multiple models on the same data	Can be confident the best-performing model is deployed.